BRAC UNIVERSITY

Quiz 1 **Department of Computer Science and Engineering**

CSE110: Programming Language I Duration: 40 minutes Marks: 20

Name: (Please write in CAPITAL LETTERS)	ID:	Section: 38

1. Write a python program that will take an integer number (N) from the user and print all the divisors of that number which are in between 2 to N **(inclusive)**. If the summation of the divisors is an odd number then divide the input number (N) by the total number of divisors and print it. If the summation of the divisors is an even number then print the input number. [Marks 10]
   . [CO 3 + CO 4]

Sample Input 1:

56

Sample output 1:

Divisors: 2 4 7 8 14 28 56

8.0

Explanation:

All the divisors of 56 in between 2 to 56 are 2,4,7,8,14,28 and 56 (Total 7 divisors). The summation of all the divisors is 2+4+7+8+14+28+56 = 119 which is an odd number. After dividing the summation by the total number of divisors, the result is 119/7 = 8.0

Sample Input 2:

9

Sample output 2:

Divisors: 3 9

9

Explanation:

All the divisors of 9 in between 2 to 9 are 3 and 9 (Total 2 divisors). The summation of all the divisors is 3 + 9 = 12 which is an even number. So the result is 9.

1. Trace the below table and write the outputs in the given boxes. You MUST show the trace table. [Marks 10] [C0 3 + CO 4]

1	`p, j , k = 0 , 20, 70`
2	`while (k > 5):`
3	`  while ( j < k ):`
4	`    p = k + j -35`
5	`    print(3 + int (p / 2))`
6	`    j += 10`
7	`  k -= 15`
8	`print(True and (False or not True) or not False)`
9	`print(j % 7 + 2)`
10	`print(k > 9 or not False)`
11	`print(p == 45)`
12	`print(14/7 + 3)`

Output